React 19 & Next.js App Router Architecture Reference Guide

Category: React and Next.js • Lead Mentor: Nasir • Licensed for Educational Study

1. React 19 Core Fundamentals & Hooks

- useState vs useReducer: Use reducers when state transitions depend on previous state or complex payloads.
- useEffect Discipline: Avoid using effects for data transformations that can be computed during render.
- Custom Hooks: Encapsulate asynchronous fetch logic, local storage sync, and window listeners into reusable hooks.

2. Next.js App Router & Server Components

- Server Components by Default: Fetch data directly on the server without client bundle overhead.
- "use client" Boundary: Push interactive state (buttons, form inputs, modal dialogs) to the leaves of the tree.
- Static Data Fetching & Caching: fetch(url, { next: { revalidate: 3600 } }) for Incremental Static Regeneration.